Equality between semi-strict and strict subgradient derivatives for C2-decomposable functions

ChatGPT (`gpt-5.6-sol`)*

Solution generated on 2026-08-27

Verifier verdict: APPROVED — progress score 3/3 (full solution)

Independent A.8 verifier assessment; pipeline details in the attribution footnote.

Abstract

We give a negative answer to the question of whether every C^2-decomposable convex function satisfies

$$\widehat{D}(\partial g)(\bar{u},\bar{y})(w) = \mathrm{cl}(D(\partial g)(\bar{u},\bar{y})(w) - K_g(\bar{u},\bar{y})^*)\,.$$

The counterexample is the support function of an elementary compact convex subset of $\mathbb{R}^3$. It satisfies the relevant nondegeneracy condition and is twice epi-differentiable at the reference point. Nevertheless, its ordinary and semi-strict graphical derivatives in a specific direction differ in precisely the way that violates the proposed identity.

1 The open problem

This paper addresses an open problem posed by Nguyen T. V. Hang, Ebrahim Sarabi in *Convergence of Augmented Lagrangian Methods for Composite Optimization Problems* [1] (Mathematics of Operations Research, 2025), Page 18, end of Section 3 (immediately after Theorem 3.9).. The website catalogue entry is problem 146 (`W4408446556_p0`) of the Open Problems in Operations Research collection.

> *Determine whether every convex function $g : Y \to \overline{\mathbb{R}}$ that is C^2-decomposable at $\bar{u} \in Y$ satisfies, for each $\bar{y} \in \partial g(\bar{u})$, the identity*
>
> $$D^b(\partial g)(\bar{u},\bar{y})(w) = \mathrm{cl}\left(D(\partial g)(\bar{u},\bar{y})(w) - K_g(\bar{u},\bar{y})^*\right) \quad \textit{for all } w \in Y.$$
>
> *Equivalently: characterize whether the semi-strict graphical derivative of the subgradient mapping ∂g can always be expressed as the (closure of the) Minkowski difference between the graphical derivative $D(\partial g)(\bar{u},\bar{y})$ and the polar cone $K_g(\bar{u},\bar{y})^*$ when g is C^2-decomposable.*

*Generated by the `base_solver_non_interactive (MOR sweep)` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260827_150058_050ee907, sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

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2 Introduction

Let $g\colon \mathbb{R}^n \to \mathbb{R}$ be convex, let $\bar{y} \in \partial g(\bar{u})$, and let

$$K_g(\bar{u}, \bar{y}) \coloneqq \{v \in \mathbb{R}^n : dg(\bar{u})(v) = \langle \bar{y}, v\rangle\}$$

denote the critical cone. We use the nonpositive polar convention

$$K^* \coloneqq \{z : \langle z, k\rangle \le 0 \text{ for every } k \in K\}.$$

The question considered here is whether the identity

$$\widehat{D}(\partial g)(\bar{u}, \bar{y})(w) = \operatorname{cl}(D(\partial g)(\bar{u}, \bar{y})(w) - K_g(\bar{u}, \bar{y})^*) \tag{1}$$

holds for every C^2-decomposable function. We construct a C^2-decomposable support function for which (1) fails.

Throughout, $\widehat{D}$ denotes the semi-strict graphical derivative.

3 Derivative conventions

Let $F\colon \mathbb{R}^n \rightrightarrows \mathbb{R}^m$ and let $(\bar{x}, \bar{y}) \in \operatorname{gph} F$. The ordinary graphical derivative is characterized sequentially as follows: $\eta \in DF(\bar{x}, \bar{y})(w)$ if and only if there exist sequences

$$t_k \downarrow 0, \qquad w_k \to w, \qquad \eta_k \to \eta$$

such that

$$\bar{y} + t_k\eta_k \in F(\bar{x} + t_k w_k) \qquad \text{for every } k.$$

The semi-strict graphical derivative is characterized analogously: $\eta \in \widehat{D}F(\bar{x}, \bar{y})(w)$ if and only if there exist sequences

$$t_k \downarrow 0, \qquad w_k \to w, \qquad a_k \in F(\bar{x}), \qquad b_k \in F(\bar{x} + t_k w_k)$$

such that

$$a_k \to \bar{y}, \qquad \frac{b_k - a_k}{t_k} \to \eta.$$

The convergence $b_k \to \bar{y}$ then follows automatically.

4 The counterexample

Define

$$C \coloneqq \left\{(p, q, r) \in \mathbb{R}^3 : p \ge 0, \quad q \ge 0, \quad p + q \le 1, \quad r^2 \le p + q\right\} \tag{2}$$

and let

$$g(x) \coloneqq \sigma_C(x) = \max_{c \in C}\langle x, c\rangle \tag{3}$$

be its support function. We take

$$(\bar{u}, \bar{y}) = (\mathbf{0}, \mathbf{0}), \qquad w = (-1, -1, 0).$$

Our main result is the following.

Theorem 1. *The function $g = \sigma_C$ is finite, continuous, convex, sublinear, and C^2-decomposable at $\mathbf{0}$. It satisfies the nondegeneracy condition*

$$\operatorname{par} \partial\vartheta(\mathbf{0}) \cap \ker \nabla\Xi(\mathbf{0})^* = \{\mathbf{0}\}$$

for the identity decomposition described below. It is also twice epi-differentiable at $(\mathbf{0}, \mathbf{0})$.

At $w = (-1, -1, 0)$, one has

$$K_g(\mathbf{0}, \mathbf{0}) = \mathbb{R}^2_- \times \{0\}, \tag{4}$$

$$D(\partial g)(\mathbf{0}, \mathbf{0})(w) = \{0\}^2 \times \mathbb{R}, \tag{5}$$

$$\widehat{D}(\partial g)(\mathbf{0}, \mathbf{0})(w) = \left\{(\alpha, \beta, \gamma) \in \mathbb{R}^3 : \alpha + \beta \leq 0\right\}. \tag{6}$$

Consequently, the proposed identity (1) fails.

5 Geometry and regularity of the example

Proposition 1. *The set C is nonempty, compact, and convex. Consequently, $g = \sigma_C$ is finite, continuous, convex, and sublinear, with*

$$\partial g(x) = \arg\max_{c \in C} \langle x, c \rangle, \qquad \partial g(\mathbf{0}) = C. \tag{7}$$

Moreover, g is C^2-decomposable at $\mathbf{0}$ and satisfies the stated nondegeneracy condition.

Proof. The set C is the intersection of the halfspaces

$$p \geq 0, \qquad q \geq 0, \qquad p + q \leq 1$$

with the sublevel set

$$\left\{(p, q, r) : r^2 - p - q \leq 0\right\}.$$

The function $(p, q, r) \mapsto r^2 - p - q$ is convex, so C is convex and closed. Furthermore,

$$0 \leq p, q \leq 1, \qquad r^2 \leq p + q \leq 1,$$

so C is bounded and therefore compact.

The standard properties of support functions now show that g is finite, continuous, convex, and sublinear. Since C is compact, the maximum in (3) is attained, and the subdifferential formula (7) follows. At $x = \mathbf{0}$, every point of C is a maximizer, so $\partial g(\mathbf{0}) = C$.

For the C^2-decomposition, take

$$\Xi = \mathrm{id}_{\mathbb{R}^3}, \qquad \vartheta = g.$$

Then

$$g(x) = g(\mathbf{0}) + \vartheta(\Xi(x)), \qquad \Xi(\mathbf{0}) = \mathbf{0},$$

and Ξ is C^∞, while ϑ is sublinear. Moreover,

$$\nabla\Xi(\mathbf{0})^* = I, \qquad \ker \nabla\Xi(\mathbf{0})^* = \{\mathbf{0}\}.$$

Therefore

$$\operatorname{par} \partial\vartheta(\mathbf{0}) \cap \ker \nabla\Xi(\mathbf{0})^* = \{\mathbf{0}\},$$

irrespective of the value of $\operatorname{par} \partial\vartheta(\mathbf{0})$. $\square$

6 Critical cone and second epi-derivative

Proposition 2. *At* $(\mathbf{0},\mathbf{0})$*,*

$$K_g(\mathbf{0},\mathbf{0}) = \mathbb{R}^2_- \times \{0\}, \qquad K_g(\mathbf{0},\mathbf{0})^* = \mathbb{R}^2_+ \times \mathbb{R}.$$

In addition,

$$d^2 g(\mathbf{0} \mid \mathbf{0}) = \delta_{K_g(\mathbf{0},\mathbf{0})}.$$

In particular, g is twice epi-differentiable at $(\mathbf{0},\mathbf{0})$*.*

Proof. Since g is sublinear,

$$dg(\mathbf{0})(v) = g(v).$$

Also, $\mathbf{0} \in C$, so $g(v) \geq 0$ for every v. Hence

$$K_g(\mathbf{0},\mathbf{0}) = \{v : g(v) = 0\}.$$

Suppose first that

$$v_1 \leq 0, \qquad v_2 \leq 0, \qquad v_3 = 0.$$

For every $(p,q,r) \in C$,

$$\langle v, (p,q,r)\rangle = pv_1 + qv_2 \leq 0.$$

Because $\mathbf{0} \in C$, the support value is therefore zero.

Conversely, if $g(v) = 0$, comparison with $(1,0,0), (0,1,0) \in C$ gives

$$v_1 \leq 0, \qquad v_2 \leq 0.$$

If $v_3 \neq 0$, choose $0 < \varepsilon \leq 1$ and set

$$c_\varepsilon = \left(\frac{\varepsilon^2}{2}, \frac{\varepsilon^2}{2}, \varepsilon \operatorname{sgn}(v_3)\right).$$

Then $c_\varepsilon \in C$, since

$$(c_\varepsilon)_1 + (c_\varepsilon)_2 = \varepsilon^2 = (c_\varepsilon)_3^2.$$

Moreover,

$$\langle v, c_\varepsilon\rangle = \frac{\varepsilon^2}{2}(v_1 + v_2) + \varepsilon|v_3| > 0$$

for all sufficiently small $\varepsilon > 0$, contradicting $g(v) = 0$. Thus $v_3 = 0$, which proves

$$K_g(\mathbf{0},\mathbf{0}) = \mathbb{R}^2_- \times \{0\}.$$

Using the nonpositive polar convention,

$$K_g(\mathbf{0},\mathbf{0})^* = \big\{z : \langle z, k\rangle \leq 0 \text{ for all } k \in \mathbb{R}^2_- \times \{0\}\big\} = \mathbb{R}^2_+ \times \mathbb{R}.$$

Finally, under the standard $\frac{1}{2}t^2$ normalization, the second-order difference quotient is

$$\Delta_t^2 g(\mathbf{0} \mid \mathbf{0})(v) = \frac{g(tv) - g(\mathbf{0}) - t\langle \mathbf{0}, v\rangle}{\frac{1}{2}t^2} = \frac{2g(v)}{t},$$

where the last equality uses positive homogeneity.

If $v \notin K_g(\mathbf{0},\mathbf{0})$, then $g(v) > 0$, and by continuity, for every $v_k \to v$ and $t_k \downarrow 0$,

$$\frac{2g(v_k)}{t_k} \longrightarrow +\infty.$$

If $v \in K_g(\mathbf{0},\mathbf{0})$, all the quotients are nonnegative, while the constant recovery sequence $v_k = v$ gives value zero. Thus the quotients epi-converge to $\delta_{K_g(\mathbf{0},\mathbf{0})}$. $\square$

7 The ordinary graphical derivative

Proposition 3. *For $w = (-1, -1, 0)$,*

$$D(\partial g)(\mathbf{0}, \mathbf{0})(w) = \{0\}^2 \times \mathbb{R}.$$

Proof. Let

$$\eta = (\eta_1, \eta_2, \eta_3) \in D(\partial g)(\mathbf{0}, \mathbf{0})(w).$$

There exist sequences

$$t_k \downarrow 0, \qquad w_k \to w, \qquad \eta_k \to \eta$$

such that

$$t_k \eta_k \in \partial g(t_k w_k).$$

Because $\partial g(t_k w_k) \subseteq C$, feasibility gives

$$(\eta_k)_1 \geq 0, \qquad (\eta_k)_2 \geq 0,$$

and consequently

$$\eta_1 \geq 0, \qquad \eta_2 \geq 0.$$

The point $t_k \eta_k$ maximizes $c \mapsto \langle t_k w_k, c \rangle$ over C. Comparing it with $\mathbf{0} \in C$ gives

$$\langle t_k w_k, t_k \eta_k \rangle \geq 0,$$

or equivalently,

$$\langle w_k, \eta_k \rangle \geq 0.$$

Passing to the limit yields

$$-\eta_1 - \eta_2 = \langle w, \eta \rangle \geq 0.$$

Together with $\eta_1, \eta_2 \geq 0$, this forces

$$\eta_1 = \eta_2 = 0.$$

Thus

$$D(\partial g)(\mathbf{0}, \mathbf{0})(w) \subseteq \{0\}^2 \times \mathbb{R}.$$

For the reverse inclusion, fix $\gamma \in \mathbb{R}$. For sufficiently small $t > 0$, define

$$c_t = \left(\frac{t^2\gamma^2}{2}, \frac{t^2\gamma^2}{2}, t\gamma \right), \qquad w_t = (-1, -1, 2t\gamma).$$

If $t^2\gamma^2 \leq 1$, then $c_t \in C$. For every $(p, q, r) \in C$,

$$\begin{aligned} \langle w_t, (p, q, r) \rangle &= -(p + q) + 2t\gamma r \\ &\leq -r^2 + 2t\gamma r \\ &= t^2\gamma^2 - (r - t\gamma)^2 \\ &\leq t^2\gamma^2. \end{aligned}$$

At c_t, equality holds. Hence

$$c_t \in \operatorname*{arg\,max}_{c \in C} \langle w_t, c \rangle = \partial g(t w_t),$$

where the last equality uses the fact that multiplying the objective by $t > 0$ does not change its maximizers.

Moreover,

$$w_t \to w, \qquad \frac{c_t}{t} = \left(\frac{t\gamma^2}{2}, \frac{t\gamma^2}{2}, \gamma\right) \longrightarrow (0, 0, \gamma).$$

Choosing any sequence $t_k \downarrow 0$ for which $t_k^2\gamma^2 \le 1$ proves

$$(0, 0, \gamma) \in D(\partial g)(\mathbf{0}, \mathbf{0})(w).$$

Since γ was arbitrary, the reverse inclusion follows. $\square$

8 The semi-strict graphical derivative

Proposition 4. *For $w = (-1, -1, 0)$,*

$$\widehat{D}(\partial g)(\mathbf{0}, \mathbf{0})(w) = \left\{(\alpha, \beta, \gamma) \in \mathbb{R}^3 : \alpha + \beta \le 0\right\}.$$

Proof. Let

$$\eta = (\alpha, \beta, \gamma) \in \widehat{D}(\partial g)(\mathbf{0}, \mathbf{0})(w).$$

By the sequential definition, there exist

$$t_k \downarrow 0, \qquad w_k \to w, \qquad a_k \in \partial g(\mathbf{0}) = C, \qquad b_k \in \partial g(t_k w_k)$$

such that

$$a_k \to \mathbf{0}, \qquad \frac{b_k - a_k}{t_k} \to \eta.$$

Since b_k maximizes $c \mapsto \langle t_k w_k, c\rangle$ over C, comparison with $a_k \in C$ gives

$$\langle t_k w_k, b_k - a_k\rangle \ge 0.$$

After division by t_k^2,

$$\left\langle w_k, \frac{b_k - a_k}{t_k}\right\rangle \ge 0.$$

Passing to the limit yields

$$-\alpha - \beta = \langle w, \eta\rangle \ge 0,$$

and therefore

$$\alpha + \beta \le 0.$$

It remains to prove the reverse inclusion. Fix

$$\eta = (\alpha, \beta, \gamma) \quad \text{with} \quad \alpha + \beta \le 0,$$

and set

$$\delta \coloneqq -(\alpha + \beta) \ge 0.$$

Define

$$\varepsilon_\gamma \coloneqq \begin{cases} 1, & \gamma \ge 0, \\ -1, & \gamma < 0. \end{cases}$$

For $s > 0$, let

$$\rho_s \coloneqq s\varepsilon_\gamma, \qquad t_s \coloneqq s^4, \qquad w_s \coloneqq (-1, -1, 2\rho_s),$$

and define

$$b_s \coloneqq \left(\frac{s^2}{2}, \frac{s^2}{2}, \rho_s \right), \qquad a_s \coloneqq b_s - t_s\eta.$$

For $s \le 1$, the point b_s belongs to C, since

$$(b_s)_1 + (b_s)_2 = s^2 = \rho_s^2 = (b_s)_3^2.$$

Furthermore, for every $(p, q, r) \in C$,

$$\begin{aligned}
\langle w_s, (p, q, r) \rangle &= -(p + q) + 2\rho_s r \\
&\le -r^2 + 2\rho_s r \\
&= \rho_s^2 - (r - \rho_s)^2 \\
&\le \rho_s^2 = s^2.
\end{aligned}$$

Equality holds at b_s, and therefore

$$b_s \in \partial g(t_s w_s).$$

We next verify that $a_s \in C$ for all sufficiently small s. Its first two coordinates are

$$(a_s)_1 = \frac{s^2}{2} - s^4\alpha, \qquad (a_s)_2 = \frac{s^2}{2} - s^4\beta,$$

and hence are nonnegative for sufficiently small s. Their sum is

$$(a_s)_1 + (a_s)_2 = s^2 - s^4(\alpha + \beta) = s^2 + s^4\delta \le 1$$

for sufficiently small s. Finally,

$$\begin{aligned}
(a_s)_1 + (a_s)_2 - (a_s)_3^2 &= s^2 + s^4\delta - \left(\rho_s - s^4\gamma\right)^2 \\
&= s^4\delta + 2s^4\rho_s\gamma - s^8\gamma^2 \\
&= s^4\left(\delta + 2s|\gamma| - s^4\gamma^2\right).
\end{aligned}$$

The last expression is nonnegative for all sufficiently small $s > 0$. Thus $a_s \in C = \partial g(\mathbf{0})$.

Now choose a sequence $s_k \downarrow 0$, sufficiently small that all the preceding feasibility conditions hold, and put

$$t_k = t_{s_k}, \qquad w_k = w_{s_k}, \qquad a_k = a_{s_k}, \qquad b_k = b_{s_k}.$$

Then

$$t_k \downarrow 0, \qquad w_k \to w, \qquad a_k \to \mathbf{0},$$

and

$$\frac{b_k - a_k}{t_k} = \eta \qquad \text{for every } k.$$

Hence

$$\eta \in \widehat{D}(\partial g)(\mathbf{0}, \mathbf{0})(w),$$

which proves the reverse inclusion. □

9 Failure of the proposed identity

By Propositions 2 and 3,

$$\begin{aligned} D(\partial g)(\mathbf{0},\mathbf{0})(w) - K_g(\mathbf{0},\mathbf{0})^* &= \big(\{0\}^2 \times \mathbb{R}\big) - \big(\mathbb{R}^2_+ \times \mathbb{R}\big) \\ &= \mathbb{R}^2_- \times \mathbb{R}. \end{aligned}$$

This set is already closed, and therefore

$$\mathrm{cl}(D(\partial g)(\mathbf{0},\mathbf{0})(w) - K_g(\mathbf{0},\mathbf{0})^*) = \mathbb{R}^2_- \times \mathbb{R}.$$

On the other hand, Proposition 4 gives

$$\widehat{D}(\partial g)(\mathbf{0},\mathbf{0})(w) = \{(\alpha,\beta,\gamma) : \alpha+\beta \le 0\}\,.$$

The vector

$$(1,-1,0)$$

belongs to the latter set, since its first two coordinates sum to zero, but it does not belong to $\mathbb{R}^2_- \times \mathbb{R}$. Consequently,

$$\widehat{D}(\partial g)(\mathbf{0},\mathbf{0})(w) \neq \mathrm{cl}(D(\partial g)(\mathbf{0},\mathbf{0})(w) - K_g(\mathbf{0},\mathbf{0})^*)\,.$$

This proves Theorem 1 and answers the question of the universal validity of (1) in the negative.

Remark (Scope of the counterexample). The example disproves only the universal validity of (1). It does not contradict the theorem preceding the question or augmented-Lagrangian convergence results established under separate sufficient hypotheses.

References

[1] Nguyen T. V. Hang, Ebrahim Sarabi. *Convergence of Augmented Lagrangian Methods for Composite Optimization Problems*. Mathematics of Operations Research, 2025. https://doi.org/10.1287/moor.2023.0324